2019 Ph H2 Q14

Section: Electricity

Topic: Monitoring and Measuring a.c.

A student investigates alternating voltage using an oscilloscope.
The waveform trace shows a sinusoidal wave with a peak-to-peak voltage of $10\,\mathrm{V}$ and a time base indicating a period of $2.0\,\mathrm{ms}$.

(a)
State the value of the peak voltage

The **peak-to-peak voltage** is the full vertical height of the wave.
The **peak voltage** is half this value:

$$V_{\text{peak}} = \frac{10}{2} = 5.0\,\mathrm{V}$$

(b)
Calculate the root mean square (r.m.s.) voltage

Use the relationship:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{5.0}{\sqrt{2}} = 3.54\,\mathrm{V}$$

(c)
Calculate the frequency of the a.c. supply

Use:

$$f = \frac{1}{T} \quad \text{where} \quad T = 2.0\,\mathrm{ms} = 2.0 \times 10^{-3}\,\mathrm{s}$$

$$f = \frac{1}{2.0 \times 10^{-3}} = 500\,\mathrm{Hz}$$

🧠 Revision Tips

- Peak-to-peak is double the **amplitude** (vertical height of one full wave)
- For sinusoidal signals:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$$

- Always convert milliseconds to seconds when calculating frequency:

$$f = \frac{1}{T}$$